

EE 538: Data-Driven Control

Course Code: EE 538

Course Name: Data-Driven Control

L-T-P-C: 3-0-2-4

Prerequisites: IC 111 Linear Algebra or similar course, EE 509 Linear Dynamical Systems, EE 301 Control Systems.

Intended for: B.Tech. (CSE, DSE, EE and ME) III/IV year/ M.Tech/M.Tech(R) /Ph.D (any branch) or instructor's consent

Distribution: Elective for B.Tech. III/IV year/ M.Tech/M.Tech(R)/PhD

Semester: Even

Approval: 61st BoA (to be reported to Senate)

Preamble

This course is meant to serve as an introduction to the emerging field of data-driven control. The relevance of the course in the current BTech curriculum is twofold: (i) it acquaints EE and ME students with new methods of identification and design of complex systems based on data, (ii) it exposes students of CSE and DSE to the possibility of applying data science concepts to solve engineering problems. The course could also serve as a starting point for research scholars looking for tools to study complex systems, or for those interested in developing learning algorithms for the design of such systems. It requires concepts of linear algebra and basic control systems, but the instructor may decide to cover the relevant concepts as and when required.

Course Modules

- **Data-driven modelling for linear systems:** Introduction to system identification and model reduction. Linear system identification from measurement data. Fundamental lemma, Persistency of Excitation (PE), Eigensystem Realization Algorithm. Balanced Truncation and Balanced Proper Orthogonal Decomposition. Models: Auto Regressive (AR), Moving Average (MA), ARMA. Kernel, image, latent variables. (10 lectures)
- **Dynamic Mode Decomposition (DMD):** Formulation, DMD algorithm, System identification with DMD, DMD with control. (6 lectures)
- **Virtual Reference Feedback Tuning (VRFT):** Methodology and design, measurement noise effect, non-minimum phase plants, multivariable plants, closed-loop stability, iterative feedback tuning (IFT). (6 lectures)
- **Data-driven control:** Controllability and stabilizability from data, noisy data, stabilization using data, data-driven LQR, h_2 , h_∞ control, model predictive control, DeePC. (8 lectures)
- **Data-driven modelling for nonlinear systems:** Nonlinear system identification from measurement data. Nonlinear Auto Regressive Moving Average model with Exogenous input (NARMAX) methods. Sparse Identification of Nonlinear Dynamics (SINDy). Koopman theory. Koopman operator based linearization. (8 lectures)

Lab Experiments

The lab sessions are to be conducted on 2-hour slots, aligned with the lectures so the concepts needed for the lab experiments are already discussed in the theory class. The lab experiments are majorly based on the tools available in Matlab.

1. System identification using data for a practical system like a dc motor
2. Data collection and persistence of excitation (PE)
3. Subspace identification and AR, MA, ARMA
4. Tune PID controller using Virtual Reference Feedback Tuning (VRFT)
5. Iterative feedback tuning (IFT) for dc motor speed control
6. Data-driven LQR design
7. DeePC implementation for linear time invariant systems
8. Koopman based linearization of nonlinear systems
9. Data-driven control of Furuta pendulum
10. Data-driven control of Maglev systems

Textbooks and papers

1. S.L. Brunton and J.N. Kutz, *Data-driven science and engineering: Machine learning, dynamical systems and control*, Cambridge University Press, 2019.
2. H. J. Van Waarde, M. K. Camlibel, and H. L. Trentelman, *Data-based linear systems and control theory*, ed. 1, Kindle Direct Publishing, 2025, ISBN 9798289885807
3. I. Markovsky, *Low-Rank Approximation: Algorithms, Implementation, Applications*, Springer, 2019.
4. J.N. Kutz, S.L. Brunton, B.W. Brunton and J.L. Proctor, *Dynamic Mode Decomposition: Data-Driven Modeling of Complex Systems*, Society for Industrial and Applied Mathematics, 2016.
5. C. de Persis and P. Tesi, “Formulas for Data-Driven Control: Stabilization, Optimality, and Robustness”, *IEEE Transactions on Automatic Control*, 65:3, pp 909-924, 2020.
6. H. J. van Waarde, J. Eising, H. L. Trentelman and M. K. Camlibel, “Data Informativity: A New Perspective on Data-Driven Analysis and Control”, *IEEE Transactions on Automatic Control*, 65:11, pp 4753-4768, 2020.
7. Recent journal articles in the domain specified by the course instructor.

Reference Books

1. A. Khaki Sedigh: *An Introduction to Data-Driven Control Systems*, Wiley-IEEE Press, 2023.
2. Andrew Pressley, *Elementary Differential Geometry*, 2th Edition, Springer, 2010
3. Manfredo P. Do Carmo, *Differential Geometry of Curves and Surfaces*, 2th Edition, Dover Publications, 2016